

Q2-

- Initial state: $x_0 = 0$
- Step size: 1
- Maximum iterations: 3

Step 1: Evaluate Initial State

$$f(0) = -(0 - 3)^2 + 10 = -9 + 10 = 1$$

The neighbors are $x = -1$ and $x = 1$:

$$f(-1) = -(-1 - 3)^2 + 10 = -16 + 10 = -6$$

$$f(1) = -(1 - 3)^2 + 10 = -4 + 10 = 6$$

Since $f(1) = 6$ is the highest, move to $x = 1$.

Step 2: Evaluate New State $x = 1$

$$f(1) = 6$$

Neighbors: $x = 0$ and $x = 2$:

$$f(0) = 1 \quad (\text{already computed})$$

$$f(2) = -(2 - 3)^2 + 10 = -1 + 10 = 9$$

Since $f(2) = 9$ is the highest, move to $x = 2$.

Step 3: Evaluate New State $x = 2$

$$f(2) = 9$$

Neighbors: $x = 1$ and $x = 3$:

$$f(1) = 6 \quad (\text{already computed})$$

$$f(3) = -(3 - 3)^2 + 10 = 10$$

Since $f(3) = 10$ is the highest, move to $x = 3$.

Final Answer:

After 3 iterations, the algorithm finds $x = 3$ as the final solution.

Q3

We use the probability formula:

$$P = e^{-\Delta E/T}$$

where:

- $\Delta E = 5$ (change in energy)
- $T = 2$ (temperature)

$$P = e^{-5/2} = e^{-2.5}$$

Approximating $e^{-2.5}$:

$$e^{-2.5} \approx 0.0821$$

Final Answer: $P \approx 0.0821$ (or 8.21% probability of accepting the worse solution)

Q1 MCQS

1: Major Drawback of Hill Climbing

Correct Answer: (b) It can get stuck in local optima

Explanation: Hill Climbing is a **greedy algorithm** that always moves to a better state. However, if there is no strictly better neighboring state, it **gets stuck in local optima**. It does not have mechanisms like random restarts or probabilistic acceptance to escape local minima.

2: How Simulated Annealing Overcomes Local Optima

Correct Answer: (a) Accepting worse solutions probabilistically based on a cooling schedule

Explanation: Simulated Annealing is inspired by the **annealing process in metallurgy**. It sometimes **accepts worse solutions with a probability** given by:

$$P = e^{-\Delta E/T}$$

where ΔE is the energy difference, and T is the **temperature** (cooling factor). This helps it escape local optima and explore a broader solution space.

3: Number of States Retained in Local Beam Search

Correct Answer: (a) Beam width

Explanation: In **Local Beam Search**, we **keep only the best k states at each iteration**. This number of retained states is called the **beam width**. A larger beam width allows **more exploration** but requires more computational power.